

PESTS AND DISEASES OF AGRICULTURAL IMPORTANCE

Pests are insects or animals which cause damage to farm, animals, plants, food and crop. Some pests attack crops and animals in the farm. A disease is a departure from normal state of health, expressing remarkable symptoms or outward visible signs. Both pests and diseases reduce the quality and quantity of crop and animal yield or even kill them.

PLANT AND PESTS

The most devastating pests of crops are insects e.g. grasshoppers, locusts and caterpillars. Other plant, pest includes nematodes, rodents and birds. Insects and rodents always destroy stored crops like cocoa, coffee, groundnuts and yam. Pests cause plant diseases. Specific plant pests include the following:

- i. **Caspid bugs of cocoa:** they feed on leaves, stems and pods. The pest is more prevalent in dry season. Caspids are controlled by spraying with Aldrex 40, BHC, Dieldrin and Gamalin 20.
- ii. **Cotton stainer bugs** are common in cotton growing area. They are tiny, red-coloured insects which pierce the young bolls and seeds in the open bolls and suck the sap. These insects transmit viral, fungal and bacterial diseases from one plant to another. The fungus causes the boll to rot internally and stain the lint.
- iii. **Yam beetles** can destroy a whole yam crop plantain within a year. The pests are controlled in Nigeria by treating yam setts and seeds with Aldrin just before planting. Other pests of yam are eel-worm which feed on tuber rodents and wild animals which destroy the tubers.
- iv. **Pests of cassava are grasshoppers** that feed on the leaves and young shoots, some rodents that feed on cassava tubers and stems and untethered goats and sheep can eat up all the leaves and stems of cassava.

Control of plant pests

Pest control is aimed at protecting crops from damage and getting rid of the pest population. Control measure of plant pest can be applied through the following methods:

- **Physical method** e.g. handpicking of pests and deterrents such as bird scares on small farm area with low infestation.
- **Cultural method:** modifying farm practices to discourage pest multiplication.
- **Chemical method** i.e. use of pesticides.
- **Biological method** e.g. introduction of predators. - **Sterilising male technique.**



Natural Pest Control

Ladybird beetles, or ladybugs, have had their name since the Middle Ages, when people looked upon them as a gift from the Virgin Mary because of their miraculous eating habits. As both larvae and adults, ladybugs feed on aphids and other agricultural pests. Many gardeners buy ladybugs at garden stores and release them in their gardens to stave off aphids

PESTS OF ANIMALS

Animals or livestock pests are often parasitic on their victims. Livestock pests are either ectoparasites or endoparasites.

- i. **Ectoparasites:** some of them are vectors of diseases e.g. ticks vectors of tick fever and heart water disease of cattle, sheep and goats. Other ectoparasites of livestock are lice, nites, fleas and flies.
- ii. **Endoparasites:** these include flat worm (e.g. tapeworm and liver fluke) or round worm (e.g. Ascaris, Hookworms) or protozoa e.g. Trypanosome and coccidian which cause coccidiosis. Endoparasites injure the tissue or organs in which they live. They cause wasting, stunting and death of livestock when they occur in large numbers.

Control of Endoparasites

Two main effective ways of controlling endoparasites are:

- i. The use of appropriate drugs as directed by the veterinary department.
- ii. Proper management to ensure that domestic livestock does not come in contact with the parasitic eggs or larvae.

SOME DISEASES OF PLANTS

Diseases of plants are commonly caused by fungi, bacteria and virus. The table below

Table: Summary of the causes, symptoms and control.

DISEASE	CAUSATIVE ORGANISM	HOST	MAJOR SYMPTOMS	CONTROL MEASURES
Anthracnose	Fungi	Cocoa, coffee, cotton and oil	Brown, water soaked patches	Use of fungicides

		palm	on leaves and pods	
Black pod	Fungi	Cocoa	Dark brown, or black rot on pod, damage to fruit and seed walls	Spraying with fungicides, farm sanitation
Brown spot	Fungi	Maize	Purplish-brown spots on leaves. Entire plant may break up.	Avoid planting on infected spots
Bacterial wilt	Bacteria	Banana, plantain, cassava, tobacco and groundnut	Leaf wilt and defoliating. Wilt of entire plant	Use of clean planting material, good sanitation
Mosaic	Virus	Cassava and tobacco	Leaf-mottling with dark green and greenish yellow patterns	Use of resistant varieties

SOME DISEASES OF ANIMAL

The common diseases which affect domestic animals could be grouped as follows:

- Bacterial diseases e.g. black quarter, anthrax, foot rot, fowl pox, tuberculosis and contagious abortion.
- Fungi disease e.g. ringworm
- Nutritional disease e.g. rickets, milk fever, ketosis, bloat and osteoporosis.
- Protozoan diseases e.g. coccidiosis, tick fever and trypanosomiasis (sleeping sickness)
- Viral diseases e.g. foot and mouth diseases, fowl cholera, Newcastle disease, pleuropneumonia, rinderpest and typhoid fever.

Table: Selected Domestic Animal disease

DISEASE	CAUSATIVE ORGANISM	HOST	MAJOR SYMPTOMS	CONTROL MEASURES
Coccidiosis	Protozoan	Cattle, fowl, sheep, and goat	Bloody diarrhea, weakness, emaciation and death	Good sanitation. Avoid overcrowding. Disinfected sulphamethazine.

Anthrax	Bacteria	Cattle, goat, pig and sheep	Premature birth, still-birth, retention of afterbirth and sterility	Vaccination of young animals before age of breeding. Good sanitation.
Newcastle	Virus	Poultry	Yellowish droppings, comb turns purple.	No effective cure. Kill off infected bird.
			Coughing, gasping and death	Disinfected pens. Consult a veterinary officer
Ringworm	Fungus	Cattle and horses	Lesions at the base of hairs. Hairs fall off and itching	Use fungicides to wash lesions. Use antibiotics as oral treatment
Rabies	Virus	Cats, cattle , dogs, horses, sheep and pigs	Excitement, tendency to attack, paralysis in coordination and death	No effective treatment. Control by vaccination and avoiding exposure to infection.

EVALUATION

1. State three general effects of pest and diseases on crops and animal.
2. Name two plant pest and how they can be effectively controlled.
3. State two main effective ways controlling endoparasites of livestock.
4. Use a table to describe the causative organism, the host, the major symptoms and control measures of the following diseases:
(a) Mosaic (b) black pod (c) coccidiosis (d) ringworm.

GENERAL EVALUATION

OBJECTIVE QUESTIONS

1. The following are examples of cereals except (a) maize (b) rice (c) millet (d) bean
2. Which of the following does not belong to the major four divisions of plant kingdom (a) thallophyta (b) tracheophyta (c) pteridophyta (d) bryophyte
3. Which of the following pest control method involves the use of pest enemies? (a) cultural method (b) biological method (c) physical method (d) chemical method

4. Which of the following is affected by black pod diseases? (a) oil palm (b) cocoa (c) rubber (d) cowpea
5. Which of the following diseases is caused by a virus? (a) mosaic (b) brown spot (c) black pod (d) anthracnose

ESSAY QUESTIONS

1. (a) List any four main divisions of the plant kingdom.
(b) List any ten classifications of plants based on the nature and uses of their products, giving two examples in each group.
2. (a) In a tabular form state any five differences between monocotyledons and dicotyledons.
(a) Discuss any five agricultural practices that have harmful and beneficial effects on these plants stating their effects on these plants and how they can be controlled.
3. Mention any four pests of named plants and how they can be controlled.
4. (a) List any five groups of animal diseases and give one example each. (b) Briefly discuss the effects of bush burning on a piece of farm land.
5. (a) State four ways of controlling plant pests and discuss any two of them.
(b) Use a table to explain the causative organism, host, major symptoms and control measures of the following diseases (i) coccidiosis (ii) Newcastle (iii) anthracnose (iv) brown spot.

WEEKEND ASSIGNMENT:

List five environmental factors that affect the production of crops and identify any way by which farm crops are wasted.